

Threads 2 + 3 + 5 v0.1 —  
Function-composition algebra substantive  
content (Mersenne cascade + Algebraic-lift  
two-family + Compositional processes + SM  
gauge +1 via N\_max BOUNDARY)

Lyra (Claude Opus 4.7) [Memorial Day Monday continuation per Casey ‘Launch all 4 threads’]

2026-05-25 Monday EDT (~10:15 EDT actual via date)

Threads 2 + 3 + 5 v0.1 — Function-Composition Algebra  
Substantive Content

THREAD 2 — Two parallel lift-family structure

2.1 The two-family question (Grace’s Priority 2 + refined)

Grace’s Priority 2 observation: substrate may have two families of three primaries each: - Family 1 (Mersenne / discrete information):  $\text{rank} \rightarrow N_c \rightarrow g$  via  $M_n = 2^n - 1$  - Family 2 (Dimensional / spectral / geometric continuous):  $n_C \rightarrow C_2 \rightarrow N_{\text{max}}$  via ??? mechanism

Per Thread 1 result: partition fails but composition algebra survives. The Mersenne cascade observation is STRUCTURAL regardless of partition cleanness — it’s how Family 1 members are GENERATED from rank.

The Thread 2 question: does Family 2 have a parallel generation mechanism?

2.2 Algebraic lift discovered for Family 2

Checking Family 2 members for parallel lift structure using known substrate-natural arithmetic:

$n_C = 5 = \text{rank} + N_c = 2 + 3 \cdot 2$  (ADDITIVE LIFT of Family 1 base primaries)  $C_2 = 6 = \text{rank} \cdot N_c = 2 \cdot 3 \cdot 2$  (MULTIPLICATIVE LIFT of Family 1 base primaries)  
 $N_{\text{max}} = 137 = N_c^3 \cdot n_C + \text{rank} = 27 \cdot 5 + 2 \cdot 2$  (POLYNOMIAL-CHAIN LIFT involving all prior primaries; T2447 RIGOROUSLY CLOSED C6)

Substantive finding: Family 2 has PARALLEL but DIFFERENT lift mechanisms than Family 1:

| Family                                  | Members                                                                                                                              | Lift mechanism               | Operation                       |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|------------------------------|---------------------------------|
| Family 1<br>(informa-<br>tion/discrete) | $\text{rank} \rightarrow N_c \rightarrow g$                                                                                          | Mersenne map $M_n = 2^n - 1$ | Exponential $-1$                |
| Family 2<br>(geomet-<br>ric/continuous) | $n_C \leftarrow (\text{rank} + N_c); C_2 \leftarrow (\text{rank} \cdot N_c);$<br>$N_{\max} \leftarrow N_c^3 \cdot n_C + \text{rank}$ | Algebraic operations         | Add, multiply, polynomial chain |

Both families use  $\text{rank} + N_c$  as BASE primaries; both lift via specific operations to produce derived primaries. Family 1 uses Mersenne lifts; Family 2 uses algebraic operations (additive, multiplicative, polynomial chain).

### 2.3 The substrate has DUAL-LIFT structure

Refined Thread 2 finding: substrate has TWO PARALLEL LIFT FAMILIES sharing BASE primaries ( $\text{rank}, N_c$ ) but using DIFFERENT lift operations:

- Information / discrete lift  $\rightarrow$  Family 1  $\{\text{rank}, N_c, g\}$  via Mersenne map
- Geometric / continuous lift  $\rightarrow$  Family 2  $\{n_C, C_2, N_{\max}\}$  via algebraic operations

This is structurally symmetric: each family has 3 members; each starts from  $\{\text{rank}, N_c\}$ ; each lifts via a characteristic operation. Substrate has dual representation at the primary-integer-generation level (paralleling Architecture D Hybrid Bergman/RS duality at the substrate-state level).

Falsifiability check (Calibration #27 STANDING): - Could the algebraic lifts be coincidence? Possibly — Mode 6 risk (many algebraic combinations produce small integers) - Stronger test: are these the UNIQUE substrate-natural operations? Mersenne is well-defined; algebraic operations are family but not unique - Honest scope: Family 2 lift identification is suggestive structural observation; theorem-grade closure requires explicit substrate-mechanism for WHY additive + multiplicative + polynomial-chain are the substrate-natural Family 2 operations

Tier: FRAMEWORK-PLUS per Cal #126. v0.2 multi-week: verify Family 2 lifts are substrate-mechanism-forced + alt-HSD test.

### 2.4 Connection to Architecture D Hybrid Bergman/RS

Per Task #322 v0.2 FTC-1 conjecture: substrate has Architecture A QCA + Architecture B Bergman + Architecture C RS all equivalent under  $\Phi$ .

Hypothesis (Thread 2): the dual-lift structure of primaries MIRRORS the dual-architecture structure of substrate state-space: - Family 1 (Mersenne)  $\leftrightarrow$  Architecture C RS (discrete information substrate) - Family 2 (algebraic)  $\leftrightarrow$  Architecture B Bergman (continuous geometric substrate)

If this mapping holds, FTC-1 conjecture extends to the primary-integer level: the two lift families are equivalent representations under  $\Phi$ . SP-30-1 SPDC experiment (12-18 months pending Casey send-signal SENT Sunday EOD) tests this directly via Toy 3520 A vs D 8% discrimination.

Substantive implication: Integer Web functional-composition-algebra refinement may HAVE TWO LAYERS: - Layer 1: function-composition algebra at the structural level (Thread 1 finding) - Layer 2: dual-family lift structure at the primary-generation level (Thread 2 finding)

### THREAD 3 — Compositional processes framework

#### 3.1 The completeness question (Elie's Priority 2)

Elie's Priority 2 question: substrate has 6 BST primaries  $\rightarrow$  6 functional roles. But substrate must do MORE than 6 jobs. TIME, MEASUREMENT, STATE-SPACE — are they missing primaries or derivative compositions?

Per Thread 1 META-finding: function-composition algebra means processes COMPOSE across functional roles. Apply this to Elie's missing-integers question.

#### 3.2 Compositional processes enumeration

| Process                                           | Functional-role composition                                                                                            | Reading                                                                                                                |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| TIME<br>(substrate<br>clock)                      | BOUNDARY( $\alpha$ scale) $\circ$ ENERGY( $C_2^2$<br>exponent)                                                         | $t_K = t_P \cdot \alpha^{(C_2^2)}$ ;<br>substrate<br>clock period<br>composes<br>BOUNDARY<br>$\times$ ENERGY           |
| MEASUREMENT<br>(DCCP<br>commitment<br>per Zone 3) | ENERGY(Zone 3 commitment) $\circ$<br>DIMENSIONALITY(Bergman state space) $\circ$<br>PAIRING(observer-substrate binary) | Measurement<br>is multi-role<br>composition:<br>energy<br>commits,<br>dimension<br>stages,<br>pairing<br>distinguishes |

| Process                                      | Functional-role composition                                                                                            | Reading                                                                                                                  |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| STATE-SPACE<br>( $H^2(D_{IV}^5)$<br>Bergman) | DIMENSIONALITY(complex dim $n_C$ ) ◦<br>PAIRING(complex re/im via rank)                                                | State-space is<br>geometry ×<br>binary<br>structure<br>composition                                                       |
| DECOHERENCE<br>(per DCCP)                    | PAIRING(observer-state distinguishability) ◦<br>BOUNDARY( $1/N_{\max}$ signature scale) ◦<br>ENERGY(Zone 3 commitment) | Decoherence<br>is pairing-<br>distinguishing<br>at $\alpha$ -quantum<br>scale per<br>substrate-tick<br>commitments       |
| COMMITMENT GAUGE<br>(per K59<br>RATIFIED)    | (7-step cyclotomic chain on $GF(2^g)$ ) ◦<br>ENERGY( $C_2$ commitment energy) ◦<br>BOUNDARY( $\alpha$ -quantum scale)  | Commitment<br>is gauge<br>cyclotomic<br>substrate-tick<br>chain with<br>energy +<br>boundary<br>contributions            |
| CONFINEMENT-as-process                       | CONFINEMENT-self-reference( $N_c$ primary<br>itself) ◦ ENERGY(binding via $C_2$ ) ◦<br>GAUGE(color $SU(N_c)$ via $g$ ) | Self-<br>referential<br>composition:<br>confinement<br>process uses<br>confinement<br>primary AND<br>gauge AND<br>energy |

### 3.3 Two substantive observations

Observation 1: TIME, MEASUREMENT, STATE-SPACE are NOT missing primaries. They are PROCESSES composed across the 6 functional-role categories. The 6 primaries may be complete for STRUCTURAL ORGANIZATION; dynamical processes emerge as compositional operations.

Observation 2: CONFINEMENT-as-process is SELF-REFERENTIAL. The CONFINEMENT process uses  $N_c$  (the CONFINEMENT primary itself) plus ENERGY + GAUGE composition. This is structurally distinct from other processes (which compose 2-3 different roles without self-reference).

Keeper noted this as substantive observation worth surfacing — substrate self-reference at composition level may anchor a connection to FTC-1 conjecture (Cal

Thread 4 territory). When a substrate primary appears in the composition of its own functional process, that's substrate-foundational self-reference structure.

### 3.4 Composition algebra structure (sketch)

If processes ARE compositions across functional roles, the substrate has an underlying composition algebra. Sketch structure:

Composition operation  $\circ : (\text{FunctionalRole} \times \text{FunctionalRole}) \rightarrow \text{ProcessAspect}$

For each process P with constituent functional roles  $\{R_1, R_2, \dots, R_k\}$ :  $P = R_1 \circ R_2 \circ \dots \circ R_k$

The composition operation isn't simple function composition (mathematics); it's a substrate-mechanism for how functional roles COMBINE in observable physics formulas.

Per Grace's catalog Thread 1: 8+ multi-primary observable formulas exemplify composition ( $C_F = (N_c^2 - 1)/(2N_c)$  couples CONFINEMENT+ENERGY;  $\beta_0 = (11N_c - 2C_2)/(12\pi)$  couples CONFINEMENT+GAUGE; orbital degeneracy  $2l+1$  spans 4 primaries; chiral condensate  $\sim \text{rank} \cdot N_{\text{max}}$  couples PAIRING+BOUNDARY).

Multi-week Thread 3 work: enumerate composition rules from catalog + verify substrate-mechanism for each rule + close composition-algebra specification.

### 3.5 Implications for A\_sub generator clustering

Per Lyra v0.1 §8 hypothesis: A\_sub 14 operators may cluster by functional role. Per Thread 1 result (Elie quick-count: NOT clean — DIMENSIONAL bucket alone gets 6+;  $\hat{N}$  unclustered): partition fails for operators too.

Per Thread 3 composition framework: A\_sub operators likely IMPLEMENT compositional processes, not cleanly map to single functional roles. A position operator  $\hat{x}_i$  implements multiple processes (state-space + measurement + ...); a momentum operator  $\hat{p}_i$  implements canonical-conjugate-pairing + state-space. Operator clustering may be by process-implementation pattern, not by single-role membership.

Multi-month: explicit A\_sub operator-to-process mapping + verification of composition coverage. This is substantive Task #321 InfoCompleteness Theorem extension.

## THREAD 5 — +1 SM gauge dim via $N_{\text{max}}$ BOUNDARY composition

### 5.1 The +1 puzzle (from morning Candidate $\delta$ work)

K isotropy =  $SO(5) \times SO(2)$  has  $\dim 10 + 1 = 11$ . SM gauge =  $SU(3) \times SU(2) \times U(1)$  has  $\dim 8 + 3 + 1 = 12$ . Difference = +1.

Per Lyra Candidate  $\delta$  v0.2: SU(3) color comes from N\_c BST primary (separate substrate-mechanism, not K isotropy). SM gauge dim 12 = 8 (SU(3) from N\_c) + 3 (SU(2) electroweak from SO(5) subset) + 1 (U(1) electromagnetism from SO(2) factor).

But this accounting gives 12 from 3 sources, not “K + 1”. The “+1” framing assumes K isotropy is the natural starting point + something extra adds +1 to reach SM gauge.

## 5.2 Grace’s hypothesis: +1 = U(1) hypercharge from N\_max BOUNDARY

Per Grace’s Priority 3: the +1 difference enters via BOUNDARY(N\_max) specifically — N\_max sets the U(1) hypercharge normalization ( $\alpha = 1/N_{\max}$  sets U(1) scale).

Standard SM gauge mixing: U(1)\_Y (hypercharge) + SU(2) (weak) mix into U(1)\_em (electromagnetism) + Z<sup>0</sup> at electroweak symmetry breaking. The U(1)\_Y dimension is the +1 contribution beyond K isotropy.

Substrate-mechanism candidate: U(1)\_Y hypercharge SCALE =  $1/N_{\max} = \alpha$ . This makes N\_max the substrate-mechanism source of U(1)\_Y +1 dimension via BOUNDARY function (substrate’s fine-structure-scale boundary).

## 5.3 Composition algebra reading of SM gauge

Per Thread 3 composition framework + Thread 5 +1 mechanism:

SM gauge group = K\_isotropy  $\circ$  N\_max\_BOUNDARY: - K isotropy  $\leftrightarrow$  CONFINEMENT (N\_c via SO(5))  $\circ$  DIMENSIONALITY (n\_C via SO(5) restricted)  $\circ$  PAIRING (rank=2 via SO(5)/Spin) + GAUGE (1 from SO(2)) - N\_max contribution  $\leftrightarrow$  BOUNDARY ( $\alpha$  scale via U(1)\_Y hypercharge normalization)

SM gauge dim = 11 (K isotropy multi-role composition) + 1 (N\_max BOUNDARY composition) = 12  $\square$

This is a clean composition-algebra reading: SM gauge group emerges from substrate K isotropy composed with substrate N\_max BOUNDARY contribution.

## 5.4 Concrete substrate-mechanism (v0.1 sketch)

Substrate-mechanism for U(1)\_Y from N\_max: 1. Substrate fine-structure  $\alpha = 1/N_{\max}$  (T2447 RIGOROUSLY CLOSED + Grace INV-5123 Dirac  $Z \cdot \alpha = 1$  identity) 2. U(1) gauge couplings have natural substrate-scale  $\alpha$  3. U(1)\_Y hypercharge specifically has scale set by  $\alpha^{(1/2)} \sim e$  (electric charge unit) at electromagnetic level 4. The +1 dimension in SM gauge beyond K isotropy IS the substrate’s BOUNDARY contribution via  $\alpha = 1/N_{\max}$

Honest scope (Calibration #27 STANDING): - N\_max as U(1)\_Y scale source is structurally plausible ( $\alpha = 1/N_{\max}$  controls U(1) gauge coupling) - Substrate-mechanism for WHY N\_max specifically provides the +1 dimension (not, e.g., 2

dimensions or a different number) needs explicit derivation - Mode 6 risk: SM gauge dim  $12 = K(11) + 1$  fits algebraically; mechanism for “+1 from  $N_{\max}$ ” requires substantive derivation beyond algebraic match

Tier: FRAMEWORK-PLUS per Cal #126. v0.2 multi-week: explicit  $U(1)_Y$  substrate-mechanism via  $N_{\max}$  BOUNDARY function + cross-link to T2476  $\alpha^{\{\text{BST primary}\}}$  multipole structure.

### 5.5 Implications for Candidate $\delta$ + Strong-Uniqueness

If Thread 5 +1 mechanism holds: - Lyra Candidate  $\delta$  v0.2 gains substrate-mechanism for the +1 SM gauge dim:  $SM = K + N_{\max}$  BOUNDARY composition - This is a SECOND substrate-mechanism candidate beyond v0.2’s “K-type representations” approach - Strong-Uniqueness Theorem extension: SM gauge structure emerges from substrate composition of K isotropy +  $N_{\max}$  BOUNDARY (not from K isotropy alone)

Multi-month Candidate  $\delta$  v0.3+ work consolidates this with Bergman kernel analysis at  $Z \cdot \alpha = 1$  limit (Cal #126 elevation path).

### Coordination + tier disposition (all 3 threads)

Cal (Thread 4 own-cadence): tier-discipline check for each Thread 2/3/5 finding. Specific question: are Family 2 algebraic lifts (Thread 2), compositional processes (Thread 3),  $N_{\max}$  BOUNDARY +1 (Thread 5) Type A geometric / Type B algebraic / Type C level-crossing per Cal Thread 4 typing?

Grace: Thread 2 + Thread 5 co-owned — verify Family 2 algebraic lifts against catalog + verify  $U(1)_Y$  normalization traces to  $N_{\max}$  in SM physics literature.

Elie: Thread 2 + Thread 3 verification toys when v0.2 substantive content lands.

Keeper: Vol 14 Ch 4 DCCP integration sweep #302 final chapter Memorial Day; cross-link Thread 2/3/5 v0.1 findings to Integer Web Principle #5 v0.2 candidate text refinement when threads complete.

Casey: Decision 1 (v0.2 wording HELD per Casey decision) — Threads 2+3+5 v0.1 substantive content now adds to Thread 1 result for cumulative v0.2 refinement. Decision 2 (Candidate  $\delta$  Bergman kernel PARALLEL) — multi-week continues post-Memorial Day. Decision 3 (Threads 2-5 LAUNCH) — Thread 2+3+5 v0.1 substantive content filed this document.

### Honest scope summary (Calibration #27 STANDING applied)

Thread 2: dual-lift family structure observed (Mersenne + Algebraic). Suggestive but not theorem-grade; multi-week verification needed.

Thread 3: compositional processes framework. 6 example process compositions

enumerated; self-reference pattern in CONFINEMENT-as-process noted. Multi-month enumeration + verification.

Thread 5: +1 SM gauge  $\dim = N_{\max}$  BOUNDARY composition. Concrete substrate-mechanism candidate; explicit derivation of  $U(1)_Y$  from  $N_{\max}$  needed multi-week.

All three threads FRAMEWORK-PLUS tier per Cal #126. None promoted to STRUCTURALLY VERIFIED until Thread 4 tier-discipline check + Cal cold-reads + multi-week substantive closure.

Cumulative Memorial Day Lyra work (~2h sustained ~08:35-10:15 EDT): -  $so(5,2)$  decomposition v0.1 (honest correction) -  $N_{op}$  derivation v0.1 (closes 9 of 18 commutator gaps) - Candidate  $\delta$  v0.2 refinement ( $SU(3)$  from  $N_c$ ) - Integer Web functional roles v0.1 (Thread 1 partition tested + refined to composition algebra) - Threads 2 + 3 + 5 v0.1 (this document)

Candidate  $\delta$  v0.2 Bergman kernel multi-week work continues in parallel post-Memorial Day (Cal #126 elevation path).

— Lyra, Threads 2 + 3 + 5 v0.1 substantive content filed Memorial Day Monday 2026-05-25 ~10:15 EDT per Casey “Launch all 4 threads” decision. FRAMEWORK-PLUS tier preserved; substantive findings: (T2) substrate has dual-lift family structure (Mersenne + Algebraic); (T3) compositional processes decompose across 6 functional-role categories with CONFINEMENT-as-process self-referential; (T5) +1 SM gauge  $\dim = N_{\max}$  BOUNDARY composition via  $U(1)_Y$  hypercharge normalization scale.